SDG BLOCKCHAIN ACCELERATOR

Impact Measurement Framework

1. Project Information

- **Project Name:** Blockchain-Based Traceability & Credit Distribution System for Plastic Recovery in India
- **Challenge & UNDP Office:** Lack of traceability, informal waste picker exclusion, and unverifiable recycling impact – UNDP India

- **Document Version:** v1.0 – Pilot Impact Measurement Framework

2. SDG Alignment

(Specify which UN Sustainable Development Goals your project addresses. Include the specific targets and indicators.)

SDG 12: Responsible Consumption and Production

Target 12.5: Reduce waste generation through recycling and reuse

Indicator: Kilograms of plastic traced and recycled via blockchain

SDG 8: Decent Work and Economic Growth

Target 8.3: Promote formalization and decent work for informal workers

Indicator: Number of waste pickers digitally onboarded and receiving credits

SDG 9: Industry, Innovation and Infrastructure

Target 9.4: Upgrade infrastructure and adopt clean technologies

Indicator: Blockchain traceability system deployed and operational

SDG 10: Reduced Inequalities

Target 10.2: Promote social inclusion

Indicator: Credit distribution to informal waste pickers without smartphone dependency

3. Key Performance Indicators (KPIs)

(Define measurable indicators that show your project's impact.)

- Plastic waste traced end-to-end (kg): Pilot-specific target
- Waste pickers onboarded via QR IDs: Pilot cohort
- Credits automatically generated and distributed: 100% of verified batches
- Fraud or duplication incidents: 0
- Recycler batch verifications completed: 100%
- NGO / aggregator partners onboarded: Multiple

4. Tracking Methods

(Describe how you will monitor and collect data for KPIs.)

On-chain Metrics:

- CIP-68 NFTs per batch

- Credit allocation logs

- Immutable transaction history

Off-chain Metrics:

- NGO onboarding records

- Aggregator transaction logs

- Audit and verification reports

5. Baseline vs Projected Outcomes

(State your starting point and what you expect to achieve.)

Baseline:
No digital traceability, no automated benefit distribution

Projected:
Full batch-level traceability, automated credit distribution, formal inclusion of waste pickers

6. Monitoring & Reporting Plan

(Define who is responsible for reporting, how often, and what format.)

Frequency: Quarterly

Responsible Parties: Plastiks, UNDP India, NGO partners

Format: Blockchain records + impact dashboard + PDF reports

Tools: Plastiks platform, CIP-68 NFTs, reporting dashboards

7. Risks & Mitigation

(Optional: outline potential risks in measuring impact and how you will address them.)

- Low digital literacy: QR-based IDs, no smartphones required
- Data manipulation: Immutable blockchain records
- Adoption resistance: NGO facilitation and training
- Verification gaps: Recycler proof uploads and audits